

Feedback Cycles

Lecture 4

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Feedback Cycles in Human-Computer Interaction

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Introduction to Feedback Cycles

Feedback cycles are the way in which people interact with the world and then get feedback on the results of those interactions.

Feedback Cycle in HCI

The Ubiquity of Feedback Cycles

Gulf of Execution

Gulf of Evaluation

Seven Questions for Designing Feedback Cycles

Applications in everyday lives



The Ubiquity of Feedback Cycles

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Ubiquity of Feedback Cycles

- Feedback cycles are essential for learning and adapting behavior.
- The process: do → see result → adjust actions.
- Feedback cycles are a hallmark of intelligent behavior, as they enable individuals and systems to adapt and improve over time.
- Feedback cycles are integral to human-computer interaction (HCI) and various learning processes.
- Feedback cycles occur in various contexts, everyday activities illustrating feedback cycles: reading, driving, interaction with others



reading



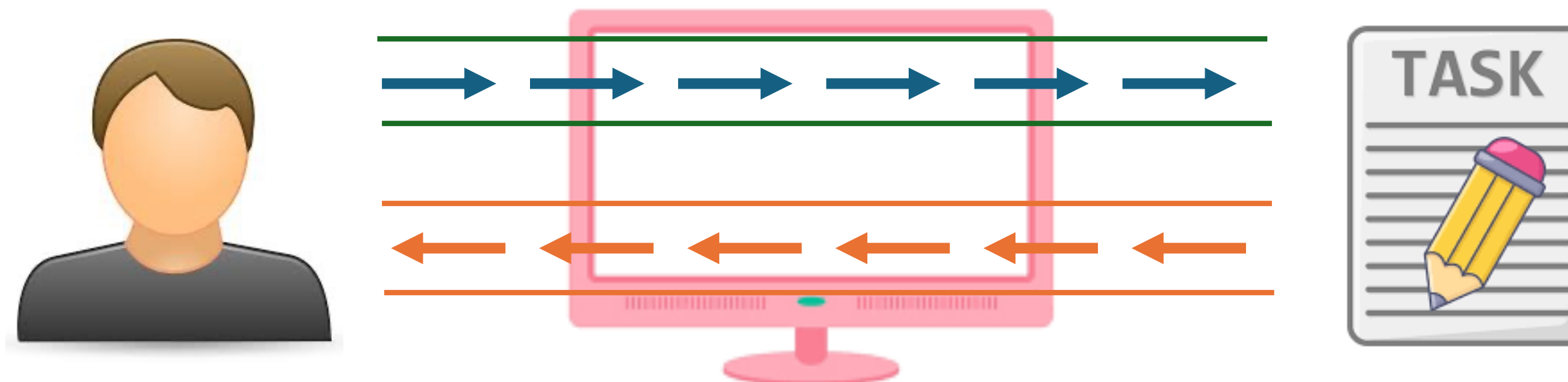
driving



interactions with others

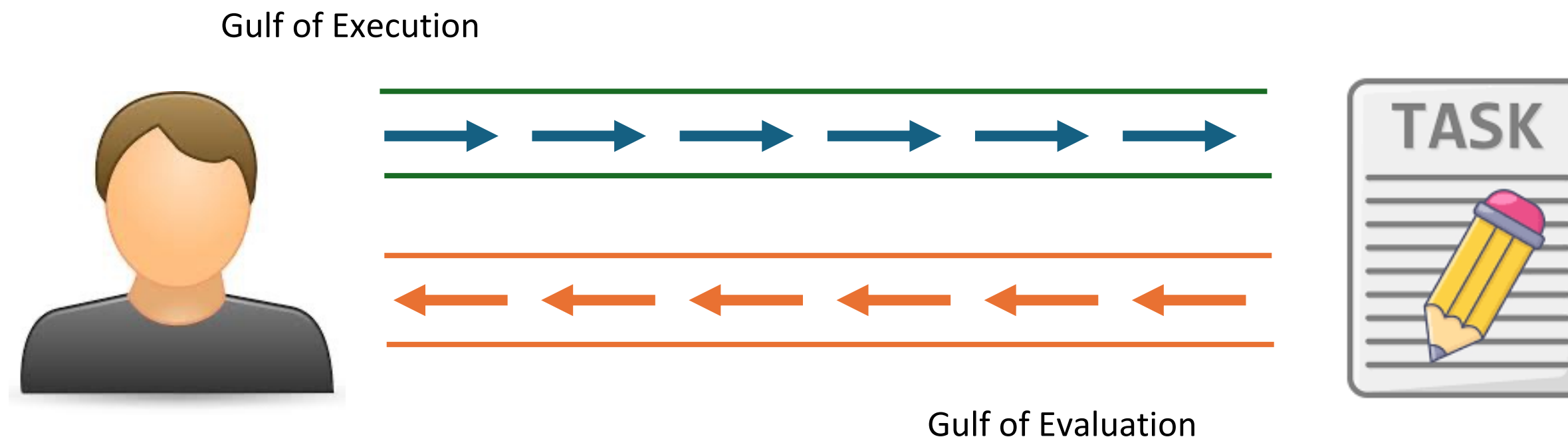
Feedback Cycle Diagram

- In the feedback cycle diagram, we have:
 - User on the left;
 - task/system on the right.
 - The user puts some input into the system through the interface and the system communicates some output back to the user again through the interface.



Challenges in Feedback Cycles

- Two general challenges in Feedback Cycles:
 - User interaction with tasks through the interface (Gulf of Execution).
 - Return of output from the task to the user via the interface (Gulf of Evaluation).





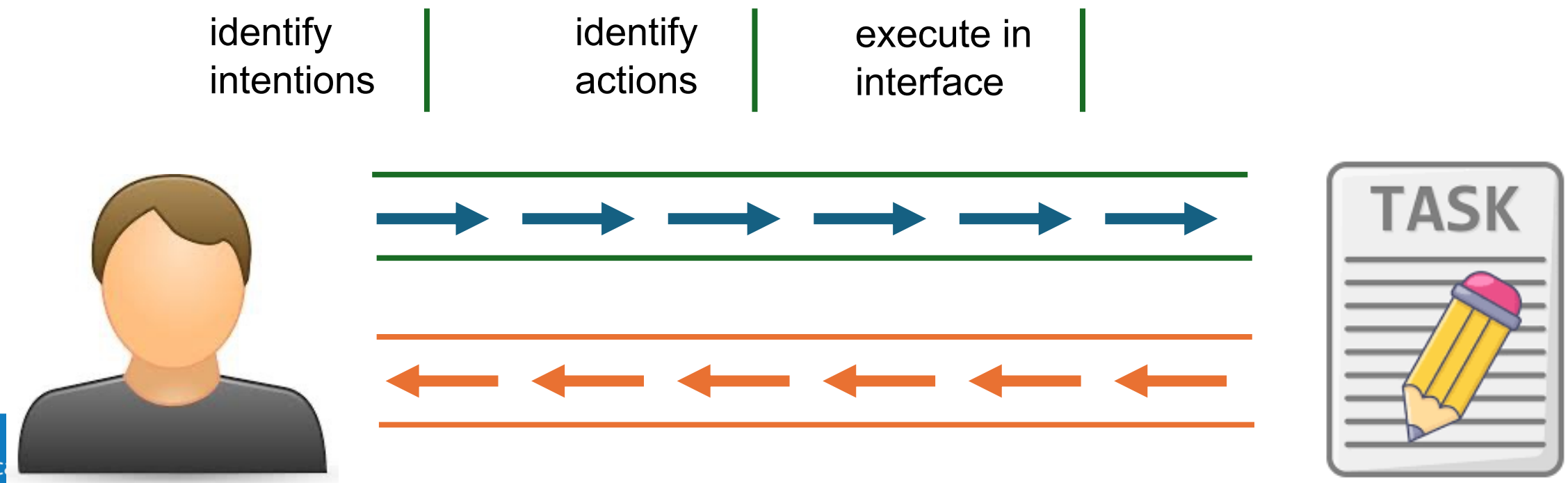
Gulf of Execution

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Gulf of Execution

- The Gulf of execution can be defined as **how do I know what I can do**.
- There are a number of components of Gulf of Execution :
 - User Goals** : Users have specific goals they want to achieve when interacting with a system. They need to identify their intentions.
 - Understanding Actions**: Users must identify the actions necessary to accomplish their goals in the context of the system.
 - Executing Action**: Once users know what actions to take, they need to execute those actions within the interface.



Problem in Gulf of Execution

- Mismatch between user understanding and system structure.
- Example: Transitioning from a VCR to On-Demand services.

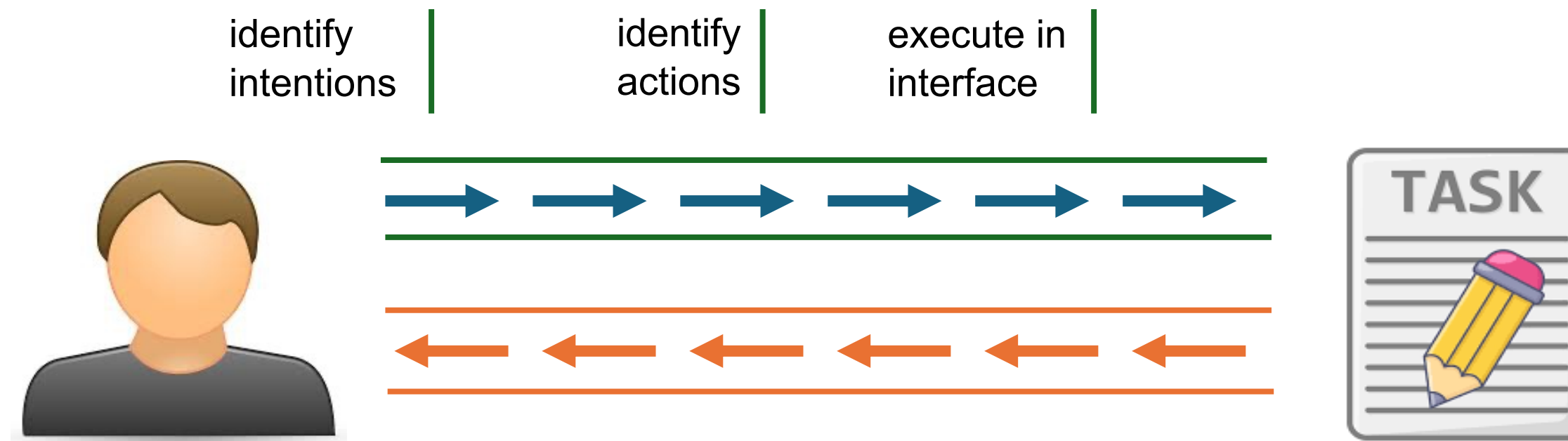


Traditional VCR



On-Demand services

Transitioning from a VCR to On-Demand services



The goal hasn't changed. They want to watch some program that's already aired. But in the context of a VCR, their intention might be to record that program. In the context of an on demand video interface, their intentions instead are to call up the existing version of that program. That's a mismatch between what they think their goal is and what their goal is in the context of this new system.

Once they know what to do, they need to actually execute that series of button presses. For example, they might know what actions to perform but they might not know where to find them. That would present a difficulty in executing those actions.

Once they understand what the goal means in their current system, they now need to know how to pull up that program. They need to know how to navigate the menus and find the program that they want to watch and then start it playing.

So, the gulf of execution takes the user from understanding their own goals to understanding their goals in the context of the system, to understanding the actions necessary to realize those goals, to actually executing those actions.

And each of these presents some difficulties.

Gulf of Execution Example

I'm making my lunch, I have a bowl of chili in the microwave.



Gulf of Execution Example (2)

- **Goal:**
 - The goal is simple: I want to heat it up.
 - Question: How hard is that?
- **Identify actions**
 - Traditional way: Cooking is defined in terms of the amount of time it takes.
 - Now: Microwave for one minute.

Gulf of Execution Example (2)

- **Actions Required (Execution)**

- Actions to achieve my goal:
 - Press Time Cook
 - Enter time (one minute)
 - Press Start

- **Complexity of Execution**

- There are many button presses just to microwave for one minute.
- Question: Could we make this better?

Simplifying Actions

- Instead of pressing Time Cook, I might just press “1” and wait.
- **The Impact of Simplification**
 - Narrowed the gulf of execution by reducing actions.
 - But could enlarge difficulty in identifying the required actions.
- **User Experience**
 - Button labels help users discover how to accomplish their goals.
 - Easier to identify actions through intuitive design.
- **Catering to Different Users**
 - This interface supports:
 - Novices: Discoverable actions.
 - Experts: Quick, simplified actions



Re-evaluating the Goal

- **Initial goal:** Heat my chili for one minute.
- **New Insight:** Was this the right thing to do?
- **Optimal Cooking**
 - After one minute, the chili might not be hot enough.
- **Feature required:** Automatic reheat function.
- **Bridging the Gulf**
 - The best bridge over the gulf of execution:
 - Encourage reframing: Heat until ready.
 - Let the microwave do the rest.



Understanding the gulf of execution can lead to better
appliance design and user experience.

Question for Audience: How can we apply this understanding
to other technologies?

Five Quick Tips for Bridging Gulfs of Execution

Tip 1 - Make Functions Discoverable

- Imagine a user in front of your interface for the first time.
- Functions should be clearly labeled and easily discoverable.

Tip 2 - Let the User Mess Around

- Encourage users to explore and discover.
- Make them feel safe by avoiding irreversible actions.

Tip 3 - Be Consistent with Other Tools

- Adopt familiar standards (e.g., Ctrl+C for copy, Ctrl+V for paste).
- Use recognizable icons like the diskette for "Save."



Five Quick Tips for Bridging Gulfs of Execution

Tip 4 - Know Your User

- Understand the components of the Gulf of Execution:
 - Identifying intentions
 - Identifying actions
 - Taking actions
- Tailor your interface for novices and experts.

Tip 5 - Feedforward

- Feedforward helps users predict outcomes of actions.
- Example: The refresh icon when pulling down the Facebook news feed.



Gulf of Evaluation

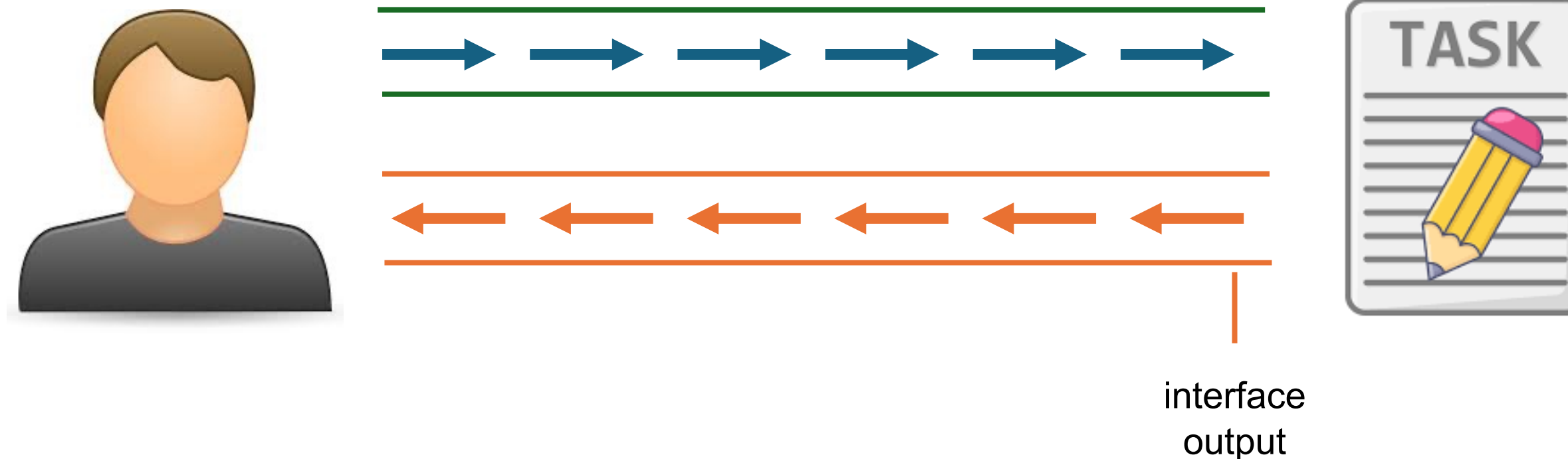
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The Gulf of Evaluation

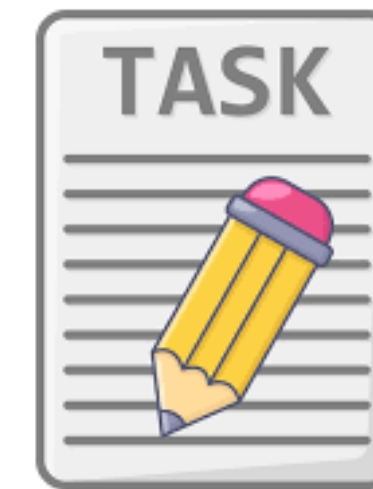
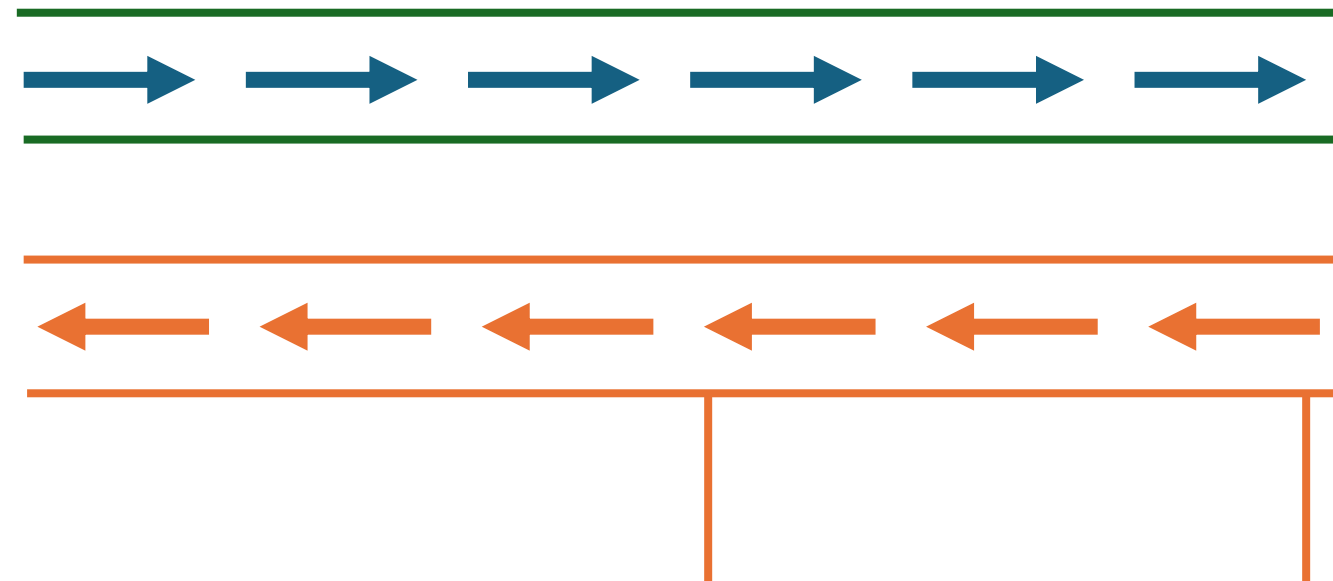
- The second challenge in Feedback cycle is how to express the output of the actions, the user have taken through the interface (Gulf of Evaluation).
- The user needs to:
 - Evaluate the new state of the system in response to their actions.
- Similar to the Gulf of Execution, the Gulf of Evaluation consists of three parts:
 1. Actual Physical Output (Interface Output)
 2. Interpretation
 3. Evaluation

Three Parts of the Gulf of Evaluation



- What does the interface actually do in response?
- Could be visual, auditory (sound), or tactile (vibration).

Three Parts of the Gulf of Evaluation

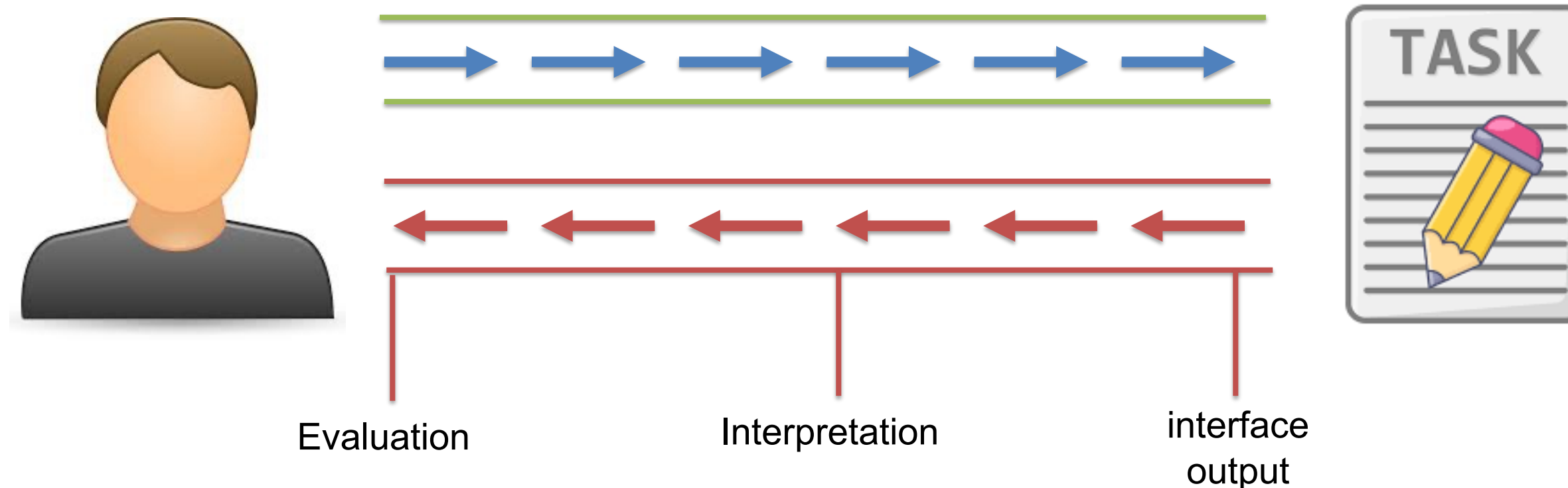


Interpretation

interface
output

- Can the user interpret the meaning of the output?
- Example: A smartphone vibrates.
- Do you pull out your phone to check the notification?

Three Parts of the Gulf of Evaluation



- Can the user evaluate if their goals were accomplished based on the output?
- Example: Submitting a form online. It might give you output that you interpret to mean that the form was received but you might not be able to evaluate whether or not the form was actually accepted.

Importance of Clarity

- Clear outputs and indications (like a buffering icon) can help users interpret system actions correctly.
- Helps users verify if their goals are being realized within the system.



Gulf of Evaluation Example: Thermostat

- **Setting a Goal**
 - Goal: Make the room warmer.
 - Action: Adjusting the thermostat.
- **System Response**
 - What does the system do?
 - The thermostat turns the heat on.
- **Evaluating System Feedback**
 - How do I know the heat was turned on?
 - Possible indicators:
 - Sound (click)
 - Physical sensation (feeling the vent)



Gulf of Evaluation Example: Thermostart

- **Issues with Indicators**

- Limitations of indicators:
 - One-time sounds might be missed.
 - Feeling heat may require accessing vents.
- Results in a large gulf of evaluation.

- **Challenges in Interpretation**

- Sound and physical feedback are not optimal.
- It requires effort to evaluate success.
- This leads to uncertainty about whether the action was successful.

- **Need for Clear Feedback**

- Ideal scenario: System provides clear feedback if unsuccessful.
- Understanding why help users adjust their actions effectively.

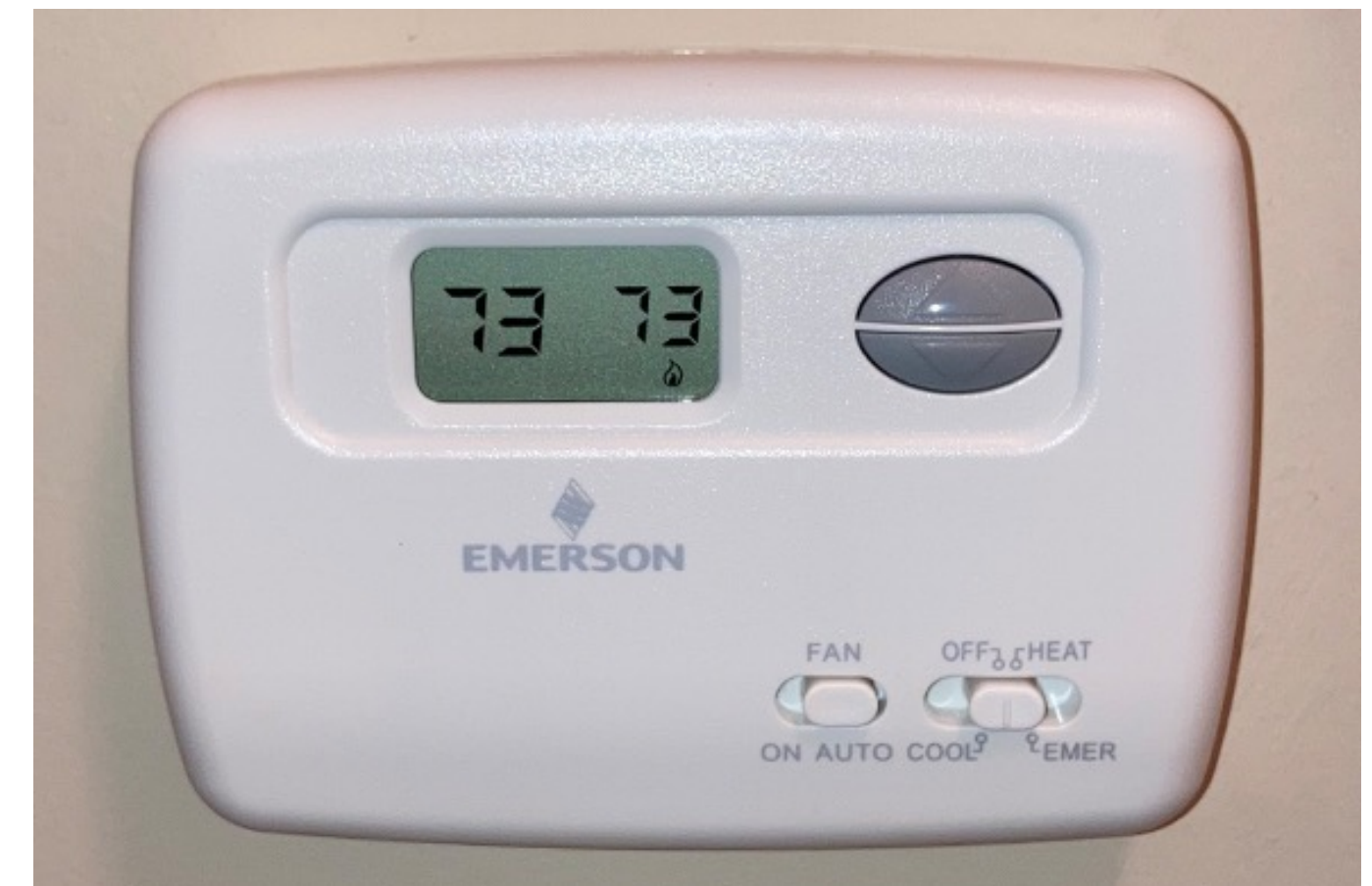
Gulf of Evaluation Example: Thermostat

- **Current Thermostat Design**

- Lack of indicators can create large gulfs of evaluation.
- Absence of direct feedback on the thermostat impacts user understanding.

- **Suggested Improvement**

- Solution: Mark on the thermostat when the heat is on.
- Improves user confidence and simplicity of evaluation.



Gulf of Evaluation Example: Thermostart

- **Limitations Remain**
 - Thermostat issues: It may activate or deactivate prematurely.
 - No indication of why certain functions aren't working.
- **Example of Confusion**
 - Evaluating temperature settings:
 - Set temperature lower than room temperature.
 - Heater status unclear.
- **Possible Indicators**
 - Potential alternative messages:
 - Showing mode settings
 - Clear indicators for malfunctions or incorrect settings

Five Quick Tips for Bridging Gulfs of Evaluation

Tip 1 - Give Feedback Constantly

- Provide feedback continuously.
- Don't wait for user actions to be fully processed.
- Indicate that input was received and help the user understand the system's status.

Tip 2 - Give Feedback Immediately

- Acknowledge input right away, even if a full response isn't ready.
- Example: Tap feedback for opening an app to reassure the user.

Tip 3 - Match Feedback to the Action

- Ensure that feedback corresponds appropriately to the magnitude of the action.
- Subtle actions receive subtle feedback, significant actions receive more pronounced feedback.

Five Quick Tips for Bridging Gulfs of Evaluation

Tip 4 - Vary Your Feedback

- Diversify feedback methods: visual, auditory, and haptic.
- Avoid overwhelming users with just visual feedback on the screen.

Tip 5 - Leverage Direct Manipulation

- Encourage direct manipulation of elements in the system.
- Intuitive actions like dragging items create a sense of interaction and engagement.

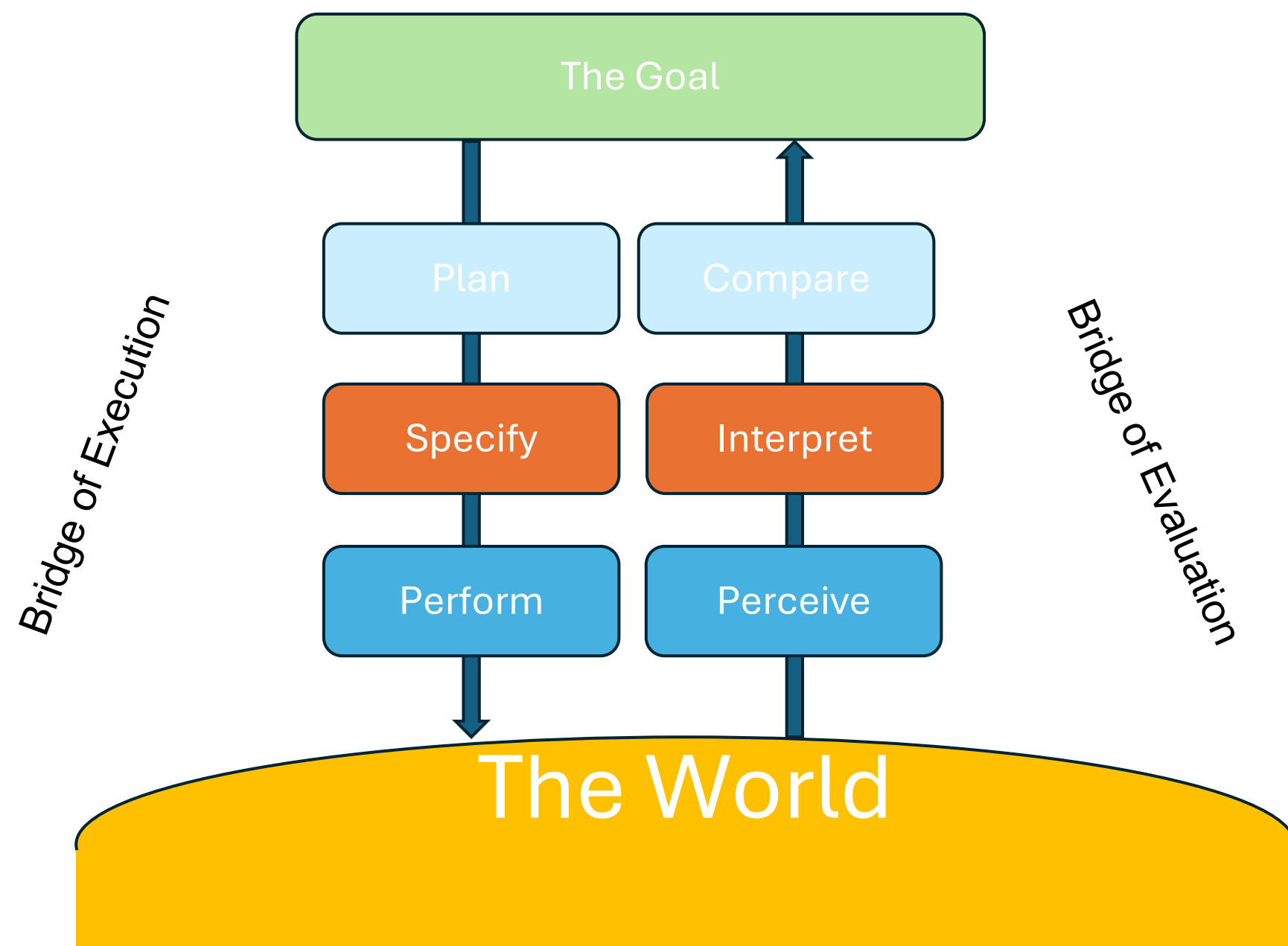


Seven Questions for Designing Feedback Cycles

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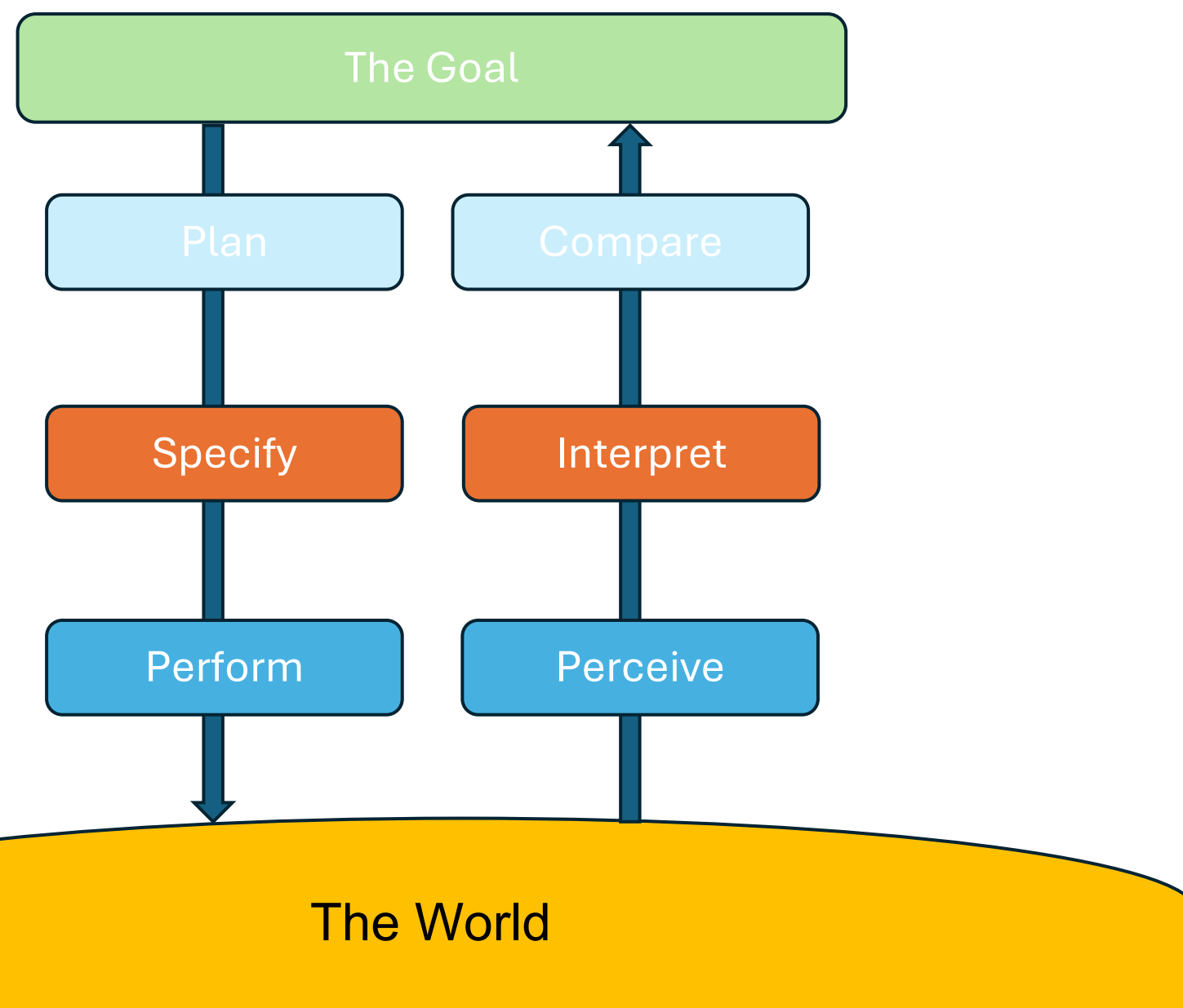
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Diagram that bridge the gulf between the Goal and the World



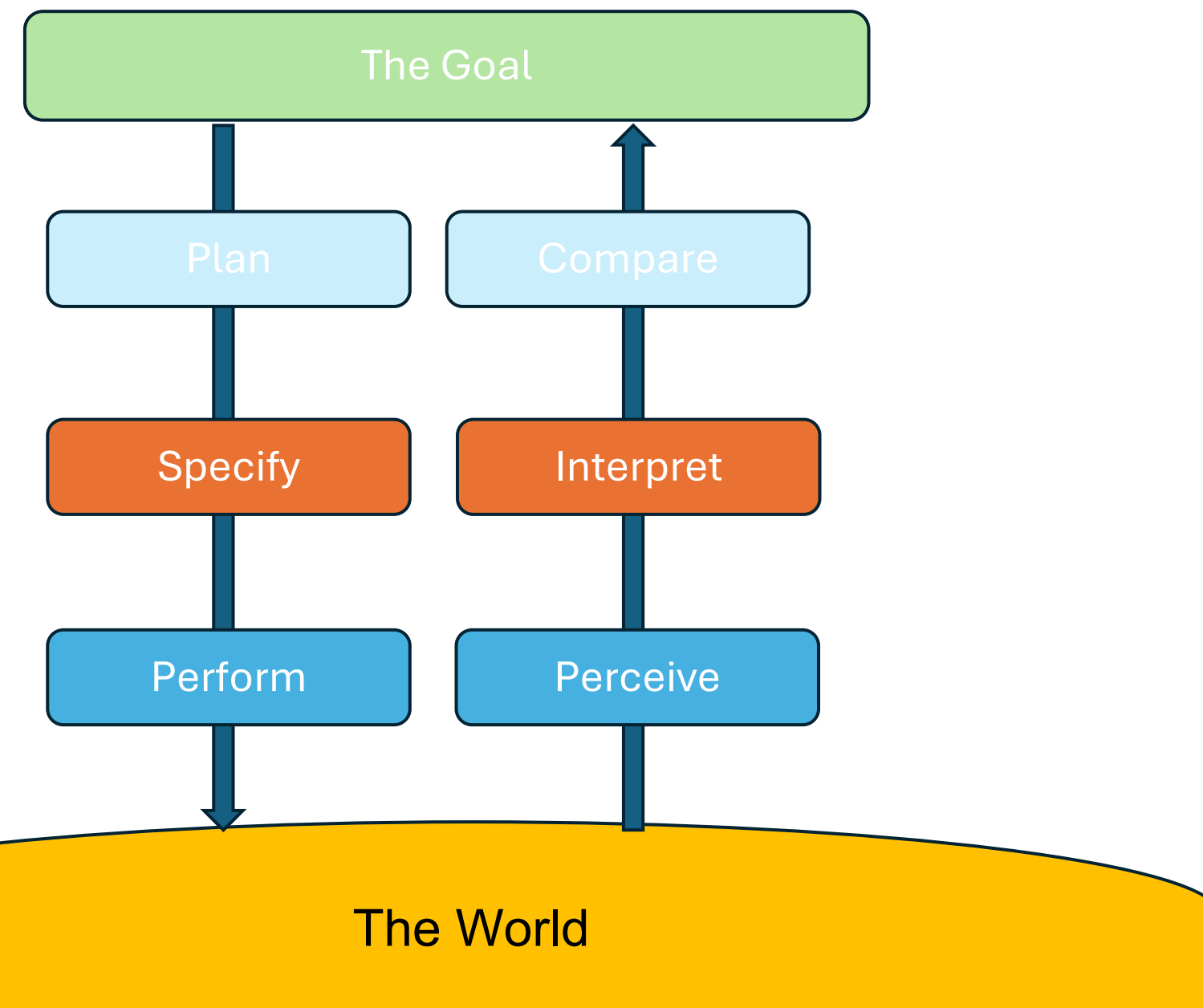
- We see a bridge of execution and a bridge of evaluation.
- These stages in that diagram show what users have to do to make the world match their goal and then confirm that the world now match their goal.

Seven Questions Introduced by Norman from These Stages



1. How easily can one tell that this device can accomplish that goal?
2. How easily can one tell what actions are possible to do with the device?
3. How easily can the user determine the mapping from their intent to the actual movement or actions they need to take with the device ?
4. How easily can the user actually perform the physical movements associated with that plan?
5. How easily can they interpret what they perceived?
6. How easily can the user determine the mapping from state to interpretation?
7. How easily can the user compare what they interpreted as happening to what they wanted to happen?

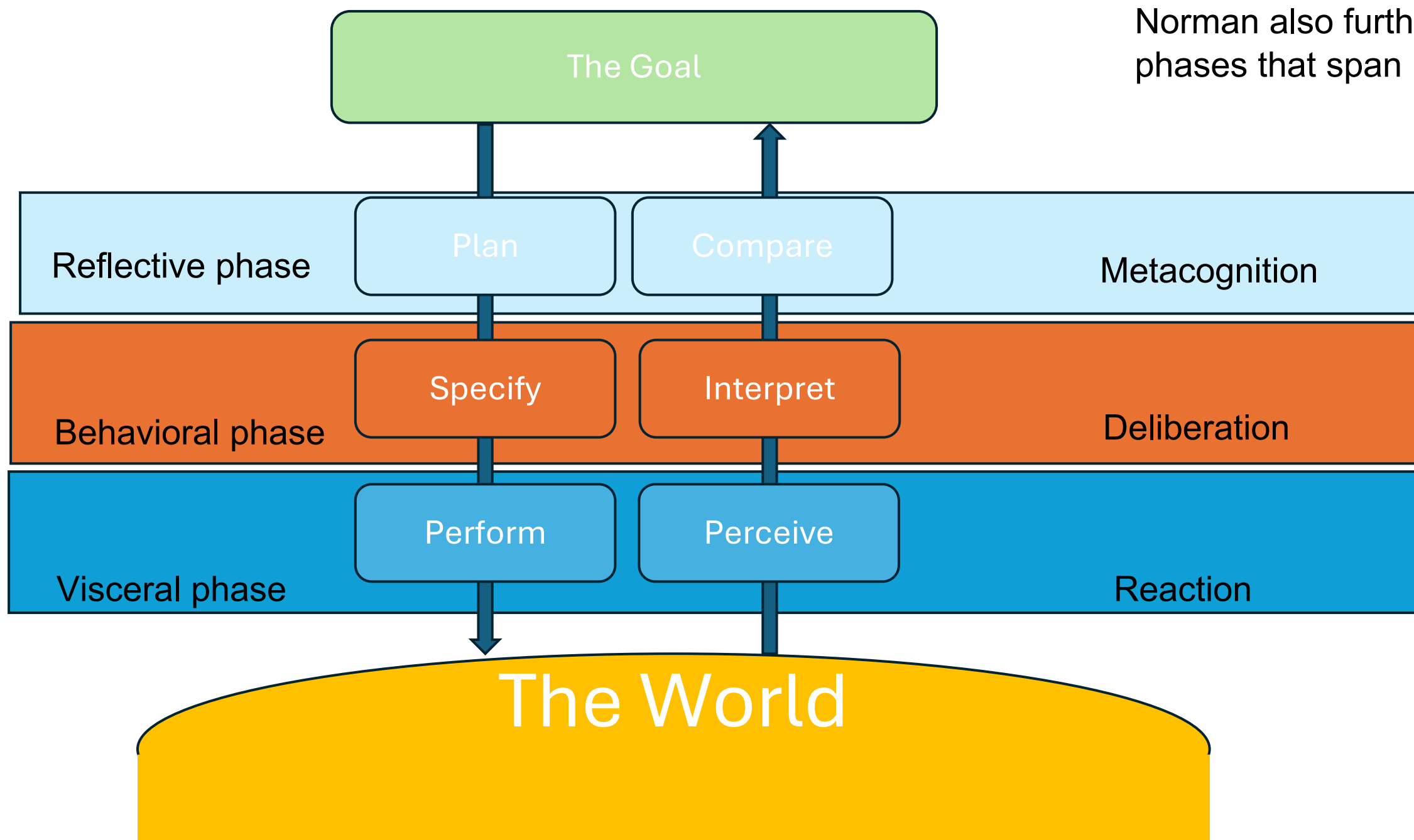
Seven Questions in Terms of What User is thinking



1. What do I want to do?
2. What are the alternatives for accomplishing it?
3. What can I do to actually perform one of those alternatives ?
4. How do I do that?
5. What happened?
6. What does that mean?
7. Is that okay?

Note: Interpreting and comparing is different. Interpreting is just about understanding what happened. Comparing involves going back to the original goal that user had

The phase from these stages



Norman also further articulates this by breaking the process into phases that span both execution and evaluation.

1. Visceral phase (closest to the world)/low level/raw reaction: the actual physical activity or the raw perceptions
2. Behavioral phase/middle level/ deliberation: what behavior to do or interpreting the outcome of these behaviors
3. Reflective phase/top level/meta cognition: thinking about the problem and planning the solution or comparing the result to our original goal



Applications in everyday lives

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Good Design

- Good design enhances user experience by providing gentle feedback.
- **Example of Good Design:**
 - A phone that quietly **clicks** every time a letter is successfully pressed.
 - This provides a subtle indication that the press has been received.

Bad Design

- Bad design overwhelms the user with unnecessary noise.
- **Example of Bad Design:**
 - A phone that loudly **shouts** every letter you type.
 - Example: "P." "I." "C."

Feedback and Actions

- **Remember:** Small actions get small feedback.
- **Example:** Typing a letter should yield subtle sounds or vibrations, not loud confirmations.
- **Special Cases for Loud Feedback**
 - The only time you might want your device to yell a confirmation is if you'd just ordered a **nuclear launch** or something similarly critical.

Example: Feedback Cycles in Modern Cars

- 15 years ago, cars were rarely discussed in the context of Human-Computer Interaction (HCI).
- Today, cars are essentially **computers on wheels**.



old car's dashboard



modern car's dashboard

Importance of Feedback Cycles in vehicles

- Feedback cycles are essential in understanding user interactions with car systems.
- Let's explore how these feedback loops apply within vehicles.
- **Traditional Ignition System**
 - Before cars had push-button starts, keys were inserted to turn on the ignition.
- **Why the Location Matters**
 - Why was the key ignition placed where it was?
 - Discussing user experience and accessibility in design.



The Evolution of Car Ignition Systems

- **Modern Ignition Systems**
 - Today, the start button can be located in various places.



The Evolution of Car Ignition Systems

- **Consistency and Expectation**

- Why do we still place it where we've always placed it?
- **Key Point:** Because that's where the driver expects it to be.

- **Bridging the Gulf of Execution**

- We help users across the gulf of execution by designing systems consistent with their expectations.

- **Intentions into Actions**

- Consistent design makes it easier for users to translate their intentions into actions.

- **When to Challenge Expectations**

- Sometimes, we might violate the principle of consistency for other benefits.

Example: Feedback Cycles in Modern Credit Card Readers

- **Context of Use**
 - My wife sells arts and crafts at local events and uses credit card readers for transactions.
- **Types of Credit Card Readers**
 - Square readers that scan credit cards using a phone.
 - One version allows swiping, while the new version allows card insertion.



Evolution of Credit Card

- **The Swipe Version**

- Swipe the card through the reader, similar to traditional card readers.
- The problem: **Lack of Feedback**
 - Typically, no feedback on whether you're swiping correctly.
 - Users can be too fast or too slow while swiping, leading to confusion.

- **Transition to New Model**

- New credit card readers allow you to simply insert the card.

- **Benefits of Insertion Model**

- Discuss how insertion models enhance feedback cycles.
- **Question for Discussion:**
 - In what ways is inserting the card better for feedback?
 - In what ways swapping and inserting models better in covid situation?





Thank You

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